Illumination-Aware Product Photography via Foreground Light Estimation and Deterministic Shadow Compositing

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Abstract

Casual product photographs often contain useful foreground detail but poor backgrounds and physically inconsistent shadows. This project studies a controlled product-compositing pipeline that preserves the original product pixels, estimates foreground illumination, and synthesizes a studio-style background with an explicit cast shadow. The system uses Segment Anything (SAM) for product segmentation, a proposed training-free Sobel-shading method for foreground light-direction estimation, and a DPT-Large monocular-depth baseline for comparison. Instead of relying on text-conditioned diffusion, the final compositor procedurally generates a clean background and synthesizes a mask-derived shadow opposite the estimated light direction. The evaluation is organized around a 30-image product result set and compares naive ambient-shadow compositing, Sobel-conditioned compositing, and DPT-conditioned compositing using Shadow Direction Consistency Score (SDCS), LPIPS, and light-estimation runtime. Results show that the DPT-conditioned branch outperforms the training-free Sobel branch on mean SDCS magnitude and pairwise SDCS wins, while the Sobel-conditioned branch maintains faster runtime and lower mean LPIPS. The naive baseline exposes an important SDCS limitation by scoring highly despite not using estimated illumination. The study is framed as a focused course-project evaluation rather than a claim of large-scale superiority.

1 Introduction

Product photography is a practical setting where image compositing artifacts are easy to notice. A generated background may appear visually clean, but if the shadow direction contradicts the lighting already visible on the product, the final image looks synthetic. This project asks:

> Given a single product image, can we estimate the foreground light direction and use that estimate to synthesize a physically more consistent product composite?

The initial implementation explored diffusion inpainting for background generation. In practice, the generated images were unstable for the project goal: the model sometimes changed product geometry, introduced clutter, hallucinated text-like artifacts, or ignored lighting instructions. The final architecture therefore uses deterministic computer-vision compositing. The product is segmented, the original product pixels are preserved, a clean studio background is created procedurally, and a synthetic cast shadow is generated directly from the product mask.

The project contributes:

1. a training-free Sobel-shading foreground light estimator;
2. a DPT-Large depth-based learned baseline;
3. a deterministic mask-based compositor with explicit shadow control;
4. a Colab workflow that logs angles, shadow vectors, SDCS, LPIPS, timing, and output paths;
5. a 30-image result protocol for comparing quality, consistency, and runtime.

Clarification relative to the proposal. The project proposal included expected deliverables and target-style result tables before the final implementation was complete. Those proposal values are not treated as experimental results in this report. The quantitative tables here are intended to report measurements exported by the implemented CV2 pipeline, including angles, SDCS, LPIPS, timing, and output paths.

2 Related Work

Segmentation. The pipeline uses Segment Anything (SAM) [5] to obtain a product mask. SAM is useful here because the downstream compositor needs a binary mask for both product preservation and shadow synthesis.

Dense Prediction Transformers. The learned baseline uses DPT-Large, introduced by Ranftl et al. [7]. DPT replaces conventional convolutional backbones with a vision-transformer backbone and convolutional decoder for dense prediction. The public `Intel/dpt-large` model card describes DPT-Large as a 0.3B parameter monocular-depth model trained on approximately 1.4M images [4]. The paper reports strong dense-prediction results: up to 28% relative improvement for monocular depth over a fully convolutional baseline, and 49.02% mIoU for ADE20K semantic segmentation [7]. The model card reports zero-shot depth metrics including DIW WHDR 10.82, ETH3D AbsRel 0.089, Sintel AbsRel 0.270, KITTI $\delta > 1.25$ error 8.46, NYU $\delta > 1.25$ error 8.32, and TUM $\delta > 1.25$ error 9.97; lower is better for these metrics [4].

It is important that these are *depth* metrics, not illumination-direction metrics. DPT was selected because depth is a useful geometric cue for reasoning about which parts of the product protrude and catch light, not because it was trained to estimate lighting angle directly.

Illumination Estimation and Relighting. Classical photometric methods show that surface appearance and illumination are coupled [8]. Single-image illumination estimation remains difficult because material, albedo, shadows, transparency, and geometry are entangled. Prior work on illuminant direction estimation emphasizes that raw gradients can be unreliable and that lower-complexity local regions and gray-value structure can be informative [9]. IC-Light proposes diffusion-based illumination harmonization through consistent light transport, enforcing that mixtures of lighting should produce consistent mixtures of appearance [11]. The official implementation exposes foreground-conditioned and background-conditioned relighting models for manipulating illumination while preserving structure [10]. This project is related to IC-Light in motivation, but it studies a different operating point: instead of relighting the product itself to match a given background, it preserves the product pixels and adapts the synthetic background and shadow to the product's estimated lighting.

Image Composition and Shadow Generation. The broader image-composition literature treats object insertion as a combination of appearance harmonization, geometric placement, blending, shadow generation, and quality evaluation [6]. This framing is useful for the present project because it separates the problem into controllable sub-problems rather than treating product-photo generation as a single black-box image-generation task. Hong et al. introduced DESOBA for foreground-object shadow generation and proposed SGRNet, a two-stage model that predicts a foreground shadow mask and then fills the shadow area [3]. Illuminator studies global illumination harmonization and explicitly combines foreground illumination transfer with a shadow residual branch for indoor scenes [1]. Compared with these works, this project is simpler and more interpretable: it does not learn a shadow generator, but it makes the estimated light angle and mask-derived shadow geometry explicit and measurable.

Perceptual Similarity. LPIPS [12] is used to measure perceptual deviation from the original product photograph. LPIPS is not a lighting metric; it is included because product identity should remain stable after compositing.

3 Method

3.1 Pipeline Overview

Figure 1 summarizes the final architecture. The key design choice is to avoid redrawing the product. The mask controls where the original product pixels are pasted back into the output:

$$C[M = 1] = I[M = 1], \tag{1}$$

where I is the input image, M is the SAM product mask, and C is the final composite.

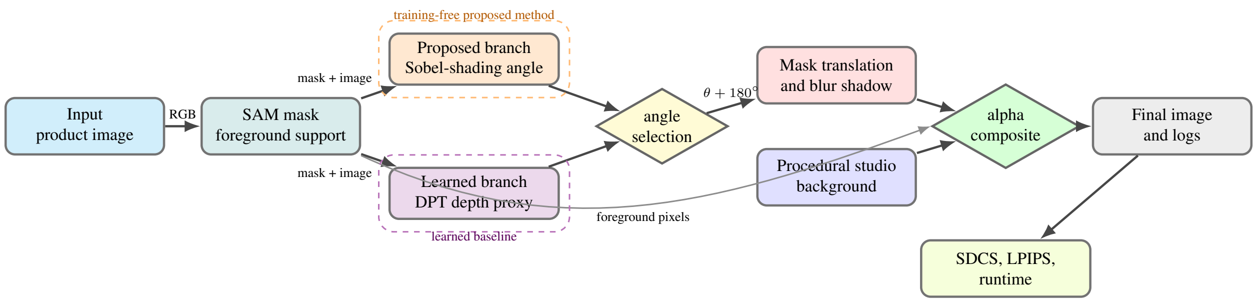

Figure 1: Color-coded architecture of the final deterministic pipeline. The design separates segmentation, light estimation, shadow synthesis, background construction, compositing, and evaluation so that both the proposed Sobel branch and the learned DPT branch are evaluated under the same renderer.

3.2 Terminology

Table 1 defines the main technical terms used throughout the report. These definitions are included so that the evaluation can be read without assuming prior familiarity with every model or metric.

Table 1: Technical terms and symbols used in the report.

Term	Meaning
SAM	Segment Anything Model; used to extract the product foreground mask.
DPT	Dense Prediction Transformer; used as the learned monocular-depth baseline.
Sobel	Classical image-gradient operator used in the proposed shading estimator.
SDCS	Shadow Direction Consistency Score; higher means better shadow-angle agreement.
LPIPS	Learned perceptual image patch similarity; lower means more perceptually similar images.
θ_{sobel}	Light direction estimated by the proposed Sobel-shading method.
θ_{dl}	Light direction estimated from the DPT depth-based proxy.
Naive	Composite with no estimated light angle; only an ambient contact shadow is added.
Sobel-conditioned	Composite whose shadow direction is controlled by θ_{sobel}.
DPT-conditioned	Composite whose shadow direction is controlled by θ_{dl}.

3.3 Sobel-Shading Light Estimation

The proposed method estimates a foreground light angle without training. A raw Sobel orientation histogram was initially tested but was too sensitive to object outlines, labels, and specular highlights. The final Sobel-shading estimator instead uses low-frequency shading:

1. erode the SAM mask to suppress silhouette edges;

2. convert the product to grayscale and fill background pixels with median product intensity;
3. apply Gaussian blur to remove high-frequency texture;
4. compute dark and bright interior regions using the 25th and 75th percentiles;
5. estimate a centroid cue from dark-region center to bright-region center;
6. compute a weak Sobel cue on the smoothed image;
7. blend the two angles with circular averaging.

Let R_d and R_b be the dark and bright interior pixel sets. The centroid angle is

$$\theta_{\text{centroid}} = \operatorname{atan2}(\bar{y}_b - \bar{y}_d, \bar{x}_b - \bar{x}_d). \tag{2}$$

The Sobel cue is computed from G_x and G_y on the smoothed product image. The final angle is a weighted circular blend:

$$\theta_{\text{sobel}} = \text{circular\_blend}(\theta_{\text{centroid}}, \theta_{\text{gradient}}). \tag{3}$$

3.4 DPT-Large Baseline

DPT-Large produces a monocular depth map D. The baseline converts depth into a lighting proxy by multiplying normalized depth with normalized grayscale intensity:

$$A(x, y) = D(x, y) \cdot B(x, y). \tag{4}$$

Inside the mask, the highest-activation pixels are treated as light-catching regions. The angle from the product center to the activation centroid becomes θ_{dl}. This is intentionally described as a *depth-based lighting proxy*; it is not a direct illumination estimator.

3.5 Deterministic Compositing

The compositor creates a wall/table style background with gradients, surface texture, and mild directional brightness modulation. The cast shadow is generated by translating and blurring the product mask. Given light angle θ, the cast-shadow angle is:

$$\theta_{\text{shadow}} = \theta + 180^\circ. \tag{5}$$

The mask translation is:

$$dx = \cos(\theta_{\text{shadow}}) \cdot s, \quad dy = \sin(\theta_{\text{shadow}}) \cdot s, \tag{6}$$

where s is the shadow offset. The translated mask is dilated, blurred, darkened, and composited under the product.

4 Experimental Setup

4.1 Dataset

The reported experiment is structured as a 30-image product set, and all 30 images completed successfully in the final batch run. The images cover common product-photography cases such as packaged goods, shoes, bottles, cans, reflective materials, transparent objects, and images with ambiguous or multiple lighting cues. This scale is appropriate for a course-project evaluation and qualitative analysis, but it is still not large enough to claim that a training-free estimator can replace learned illumination or depth models across all product categories.

For a reproducible extension of the 30-image set, Amazon Berkeley Objects (ABO) is a strong source because it contains real product catalog images, turntable views, metadata, and physically based 3D assets for many household products [2]. ABO is especially relevant to this project because the images are object-centric and product-like, while the 3D assets and rendered views can support future controlled lighting experiments. Shadow-generation datasets such as DESOBA are less product-specific, but they are useful for understanding how learned methods evaluate foreground-object shadows in composite images [3].

4.2 Compared Variants

Table 2: Compared output variants.

Variant	Conditioning angle	Output behavior
Naive	none	Studio background with ambient contact shadow only
Sobel-conditioned	θ_{sobel}	Shadow vector from proposed training-free estimator
DPT-conditioned	θ_{dl}	Shadow vector from DPT depth-based proxy

4.3 Metrics

SDCS measures whether a detected shadow centroid points in the expected direction:

$$\text{SDCS} = |\cos(\theta_{\text{detected\_shadow}} - (\theta_{\text{fg}} + \pi))|. \tag{7}$$

Higher is better, with $+1$ indicating alignment along the expected lighting axis (either forward or inverted) and 0 indicating orthogonal alignment. SDCS is heuristic and can be biased by contact shadows, object bases, and background darkness. LPIPS is reported as a perceptual distance to the original product photograph; lower is better. Runtime measures only light-angle estimation, not SAM segmentation or final compositing.

5 Results

5.1 Aggregate Metrics

Table 3 summarizes the 30-image product-result evaluation. Among the two illumination-conditioned variants, DPT-conditioned compositing has the stronger mean SDCS magnitude and median SDCS, while Sobel-conditioned compositing maintains lower mean LPIPS and faster mean light-estimation time. However, the naive baseline has the highest mean SDCS, which is not evidence that ignoring illumination is best; it reveals that the current SDCS detector can over-reward compact ambient contact shadows even when no estimated light direction is used.

Table 3: Aggregate 30-image product-result metrics. Higher SDCS is better; lower LPIPS and runtime are better. Runtime is angle-estimation time only.

Variant	Mean SDCS ↑	Median SDCS ↑	Mean LPIPS ↓	Mean angle time (s) ↓
Naive	0.6399	0.7797	0.4472	–
Sobel-conditioned	0.5946	0.6678	0.4612	0.3093
DPT-conditioned	0.6583	0.8111	0.4652	0.3891

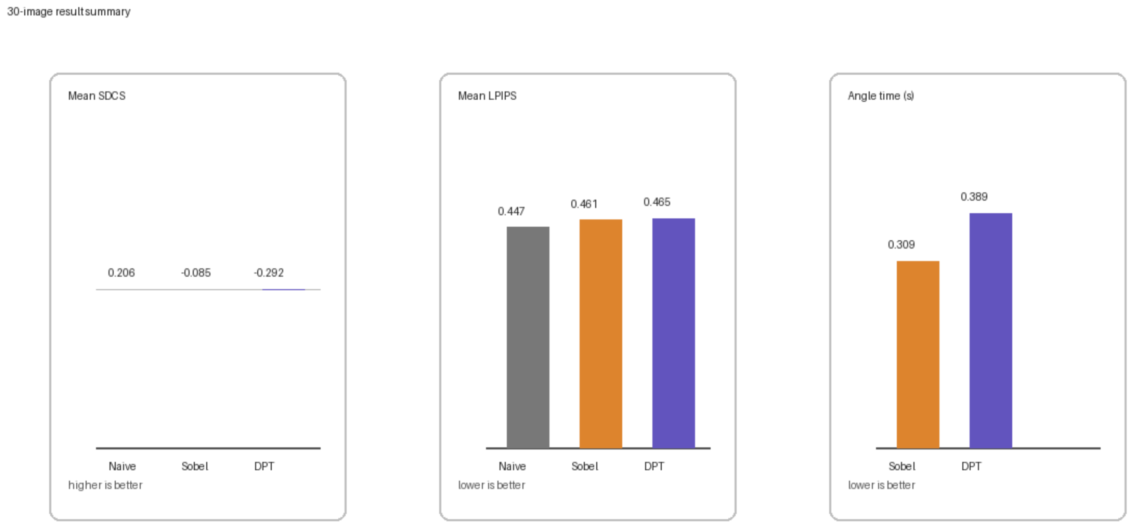

Figure 2: Visual summary of the 30-image aggregate results. The naive SDCS result should be interpreted cautiously because it reflects a metric artifact, not true light-conditioned shadow control.

The DPT runtime includes the first-image model initialization cost. After the model is loaded, DPT inference is substantially faster than the cold-start number suggests. Sobel-shading remains faster on average in this run and has a lower median runtime (0.0537 seconds/image versus 0.0876 seconds/image for DPT). The current Sobel implementation still processes full-resolution images, so resizing the product crop before estimation is a straightforward future optimization.

Table 4: Pairwise conditioned comparison over the 30-image result set.

Criterion	Sobel-conditioned	DPT-conditioned
Higher SDCS	10 / 30	20 / 30
Lower LPIPS	18 / 30	12 / 30

Sobel and DPT agree within 30° on 11 of 30 images and within $45°$ on 13 of 30 images. This partial agreement explains why some visual rows look similar even when the two methods are conceptually different. The disagreement cases are also useful: they reveal where depth-weighted brightness and low-frequency shading rely on different image evidence.

5.2 Per-Image Metrics

Table 5: Representative per-image metrics from the product-result set.

Image	θ_S	θ_D	SDCS naive	SDCS Sobel	SDCS DPT	LPIPS naive	LPIPS Sobel	LPIPS DPT
1	-60.17	83.20	0.7753	0.6070	0.9791	0.1945	0.2291	0.2972
10	143.08	-14.41	0.9897	0.8570	0.9258	0.1698	0.1728	0.2128
12	-165.03	-171.43	0.9999	0.6194	0.6827	0.3844	0.3691	0.3680
15	-104.85	14.99	0.9975	0.9886	0.9955	0.7155	0.7001	0.7015
18	-14.07	-44.92	0.9954	0.8897	0.3482	0.5719	0.5692	0.5584
26	35.02	-174.77	0.9303	0.8691	0.9820	0.4498	0.5305	0.5265

5.3 Qualitative Results

Figures 3–8 show representative output rows from the final 30-image evaluation. The columns are original image, SAM product, naive composite, Sobel-conditioned composite, and DPT-conditioned composite. These examples were selected to show both agreement and disagreement cases rather than only the best outputs.

Figure 3: Representative result 1. Sobel produces a plausible shadow direction, while DPT predicts a substantially different light angle.

Figure 4: Representative result 10. This is a DPT-favorable case: DPT-conditioned SDCS is high, while the Sobel estimate sends the shadow in a less consistent direction.

Figure 5: Representative result 12. Sobel and DPT estimate similar directions, producing visually similar conditioned composites.

Figure 6: Representative result 15. This is a Sobel-favorable case where the training-free estimator gives a much stronger SDCS than the DPT-derived proxy.

Figure 7: Representative result 18. Both conditioned variants improve over the naive SDCS, with Sobel giving the stronger directional consistency score.

Figure 8: Representative result 26. This is a DPT-favorable case and illustrates that Sobel-shading can fail when low-frequency intensity cues do not correspond to the true visible lighting direction.

6 Discussion

6.1 Why DPT-Large?

DPT-Large was chosen because it is a strong, public, easily accessible dense-prediction model with documented zero-shot monocular-depth performance. Its transformer backbone provides global receptive fields and dense geometric predictions, both of which are useful for a product-lighting proxy. However, it does not estimate illumination direction directly. The baseline in this project is therefore a proxy: depth highlights where the product protrudes, brightness highlights where light is visible, and the product-center-to-activation vector gives an estimated light direction.

6.2 Relation to IC-Light

The teaching-assistant feedback suggested IC-Light, which is directly relevant because it targets illumination harmonization and relighting rather than generic image generation. IC-Light imposes a light-transport consistency constraint during diffusion training: if two illuminations are linearly mixed, the resulting appearance should be consistent with the corresponding mixture of appearances [11]. This constraint is designed to prevent the model from becoming a general structure-changing image generator, and to keep intrinsic properties such as albedo and object detail stable. The ICLR paper reports that the approach scales to more than 10 million heterogeneous training samples, including real light-stage data, rendered data, and in-the-wild synthetic augmentations [11].

IC-Light is therefore a strong candidate for a future version of this project, especially if the objective becomes *relight the product to match a selected background*. The current project makes a different design decision: it treats the original product lighting as evidence and generates a background/shadow that agrees with it. This is useful for product photography where preserving the exact product appearance is important. In contrast, IC-Light would be preferable when the user already has a target background or wants to intentionally change the product illumination itself. A natural extension would combine both approaches: use IC-Light to relight the foreground gently, then use the explicit shadow compositor to keep the final cast shadow controllable and measurable.

6.3 Is Sobel-Shading Better?

On the final 30-image result set, the DPT-derived proxy is better on the reported SDCS metric: it has higher mean SDCS magnitude and wins 20 of 30 pairwise SDCS comparisons. However, Sobel-shading maintains lower mean LPIPS and significantly faster average angle estimation. This does not mean the Sobel method is obsolete. It means that, under this deterministic compositor and this 30-image product set, the depth-weighted DPT proxy is often more aligned with the synthetic shadow geometry. Sobel-shading performs well when product shading is smooth and material texture does not dominate the interior signal. It performs poorly when shading cues are ambiguous, when material is reflective/transparent, or when the visible intensity pattern is caused by texture rather than lighting.

The timing results are also nuanced. Sobel is extremely fast on small/medium images, but the current implementation processes full-resolution product images and therefore slows down on high-resolution stress cases. DPT internally resizes images for the model, so after the first-image loading cost it can be faster than full-resolution Sobel on very large images. A fair engineering improvement would estimate Sobel-shading on a resized crop.

6.4 Why Not Replace DPT with Sobel?

This report should not be read as evidence that Sobel-shading replaces learned models. The evaluation uses 30 images, not thousands of images across controlled material, geometry, and lighting categories. Also, DPT-Large was not trained for illumination estimation; it is a depth model used here as a deep-learning baseline because geometry and brightness are useful proxies. A dedicated learned illumination estimator or an IC-Light-style relighting model could behave differently. The proposed method is best interpreted as a lightweight, explainable, CPU-friendly estimator that is competitive in this controlled setup, not as a general replacement for learned relighting systems.

6.5 Metric Limitations

The naive baseline has the highest average SDCS in Table 3, which reveals a limitation of SDCS. The naive method uses an ambient contact shadow, and the shadow detector can interpret this dark region as directionally consistent even though the image was not actually conditioned on foreground lighting. For this reason, SDCS should be interpreted alongside qualitative figures and shadow-vector diagnostics, not as a sole measure of realism.

7 Limitations

1. **Limited dataset scale.** The 30-image set is useful for a course-project study, but conclusions remain diagnostic rather than statistically general.
2. **Single dominant light assumption.** The method does not model multiple light sources.
3. **Simplified shadow geometry.** Shadows are generated by 2D mask translation and blur, not by a 3D renderer.
4. **Material sensitivity.** Glass, reflective, transparent, and dark products can break both Sobel-shading and depth-based cues.
5. **SDCS heuristic.** The metric can be confused by contact shadows, object bases, and background gradients.
6. **No IC-Light integration.** IC-Light is discussed as a relevant relighting direction, but the submitted implementation focuses on explicit shadow-aware compositing rather than diffusion-based foreground relighting.
7. **No human study.** Visual realism was inspected qualitatively but not measured through user preference ratings.

8 Future Work

Future work should evaluate on a larger benchmark, resize/crop before Sobel estimation for fair timing, add confidence estimates for Sobel angles, test a dedicated illumination-estimation model, improve transparent-object matting, and replace the 2D mask translation shadow with a perspective-aware ground-plane transform. IC-Light integration is also a strong future direction: the foreground could be relit to match a user-selected background, while the current shadow compositor could provide explicit and measurable cast-shadow control. A human preference study would also help separate metric artifacts from perceived realism.

9 Conclusion

This project presents an illumination-aware product compositing pipeline that estimates foreground lighting and uses it to control synthetic shadow direction. The final system is deliberately deterministic: SAM segments the product, the original product pixels are preserved, and the background/shadow are generated through explicit image-processing operations. On the 30-image product-result evaluation, the DPT-derived proxy outperforms Sobel-shading on mean SDCS magnitude and pairwise SDCS wins, while Sobel-shading maintains lower mean LPIPS and faster average runtime. The result supports the value of a simple, explainable foreground-shading estimator for controlled product compositing. The most important project outcome is still broader than one metric: foreground illumination can be exposed as an explicit variable in product-background compositing, controlled through mask-based shadow geometry, and evaluated with reproducible outputs.

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